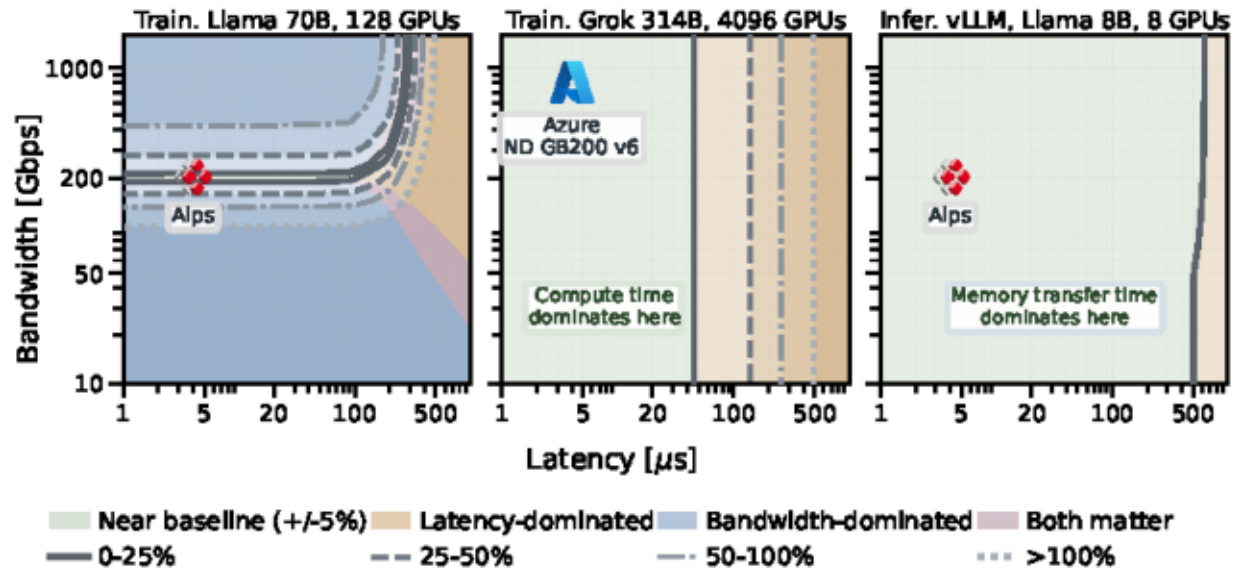


Figure 1: workload sensitivity maps

Paper plot



Artifact output: compact-data redraw

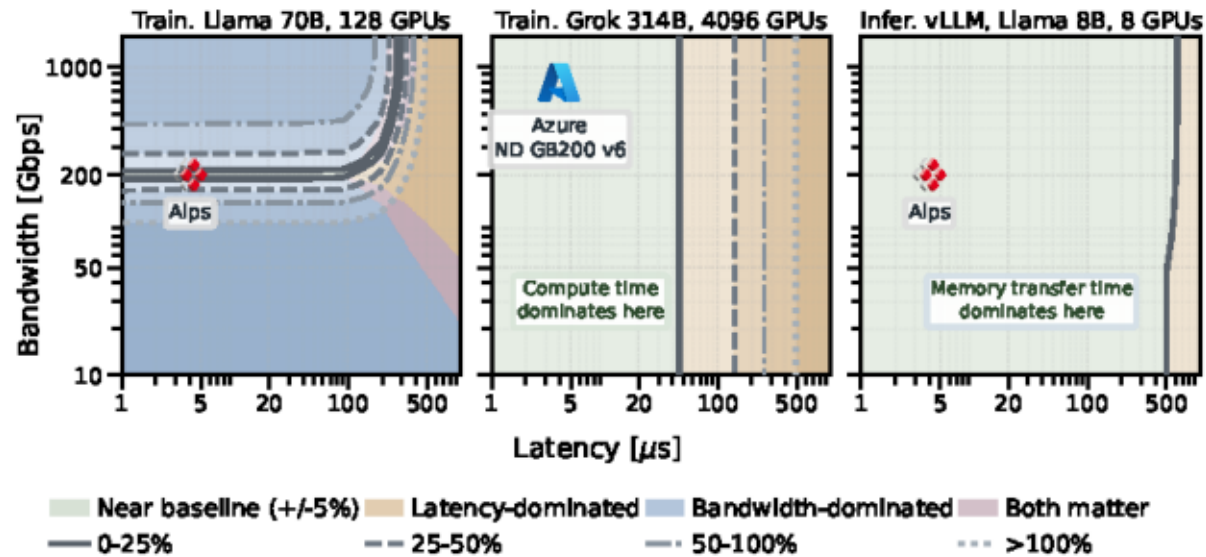
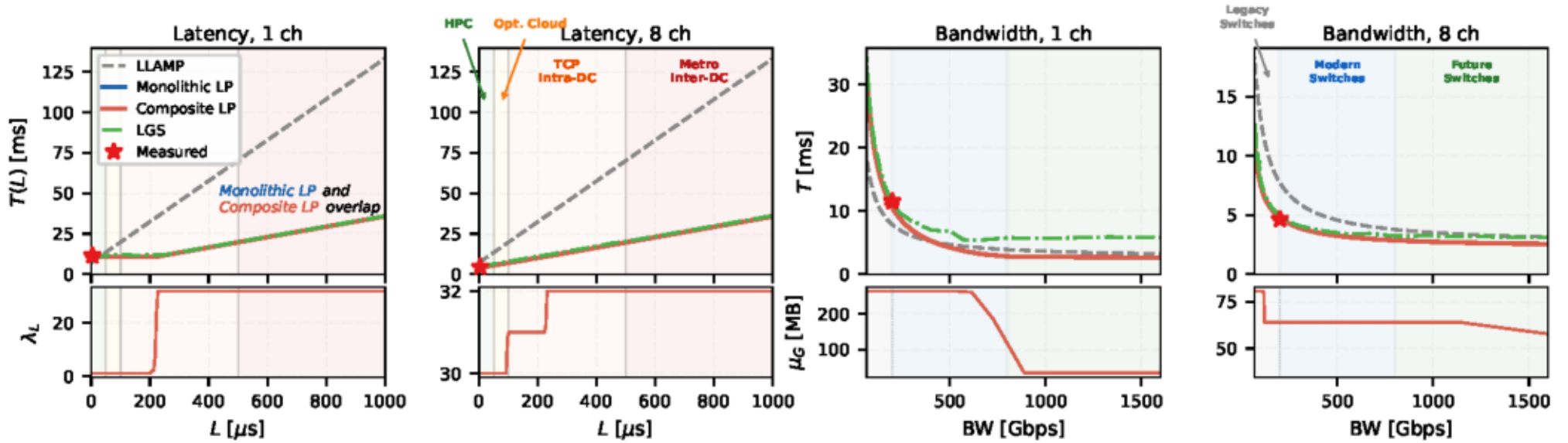
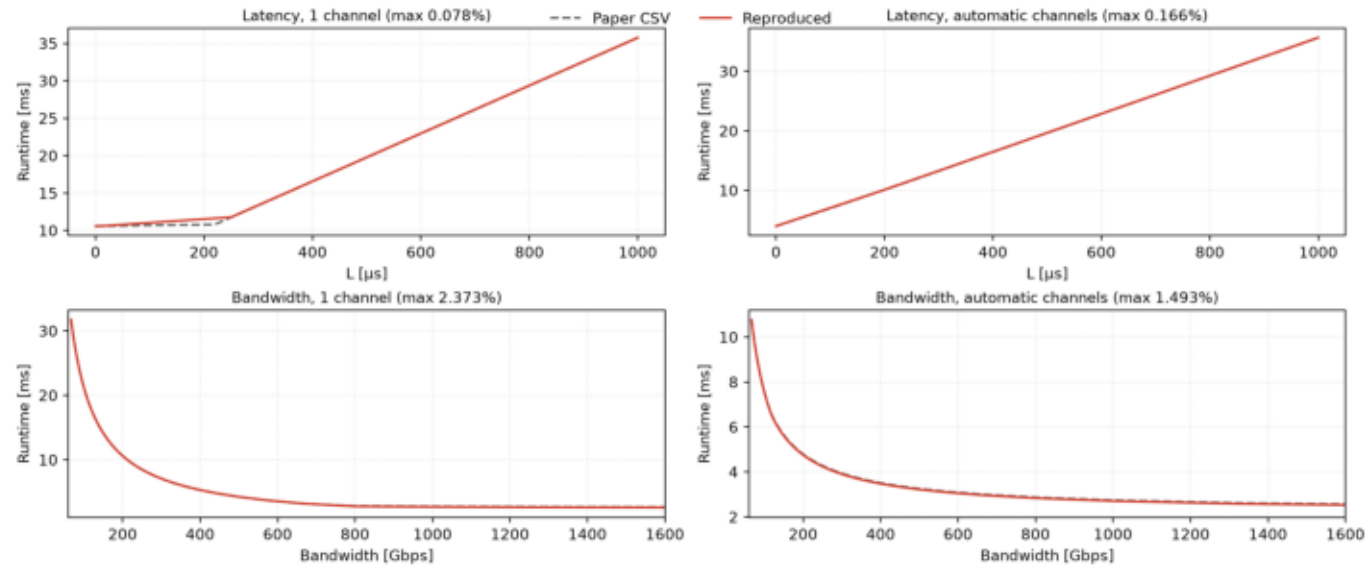


Figure 3: 128 MiB AllReduce

Paper plot



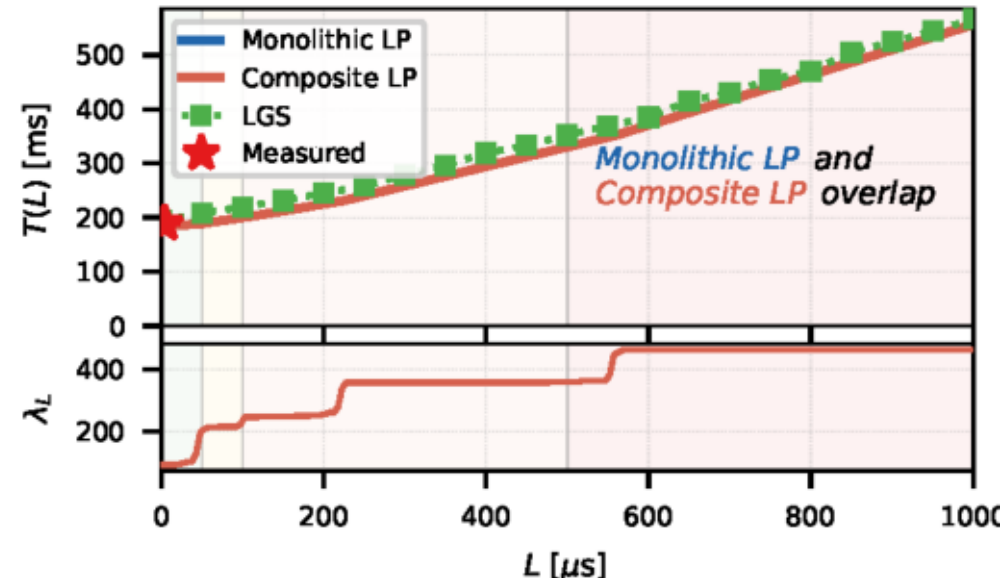
Raw-derived result: selected final Alps curves



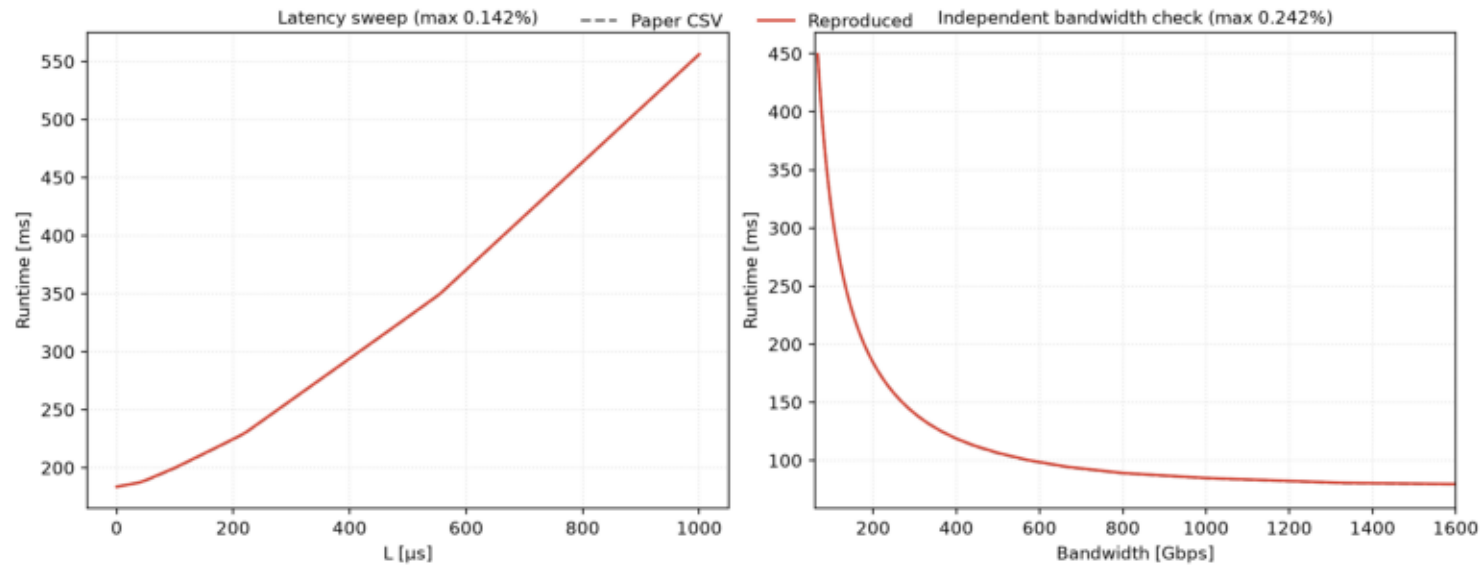
Maximum relative errors: 0.078% and 0.166% for latency; 2.373% and 1.493% for bandwidth.

Figure 4: mixed collectives

Paper plot



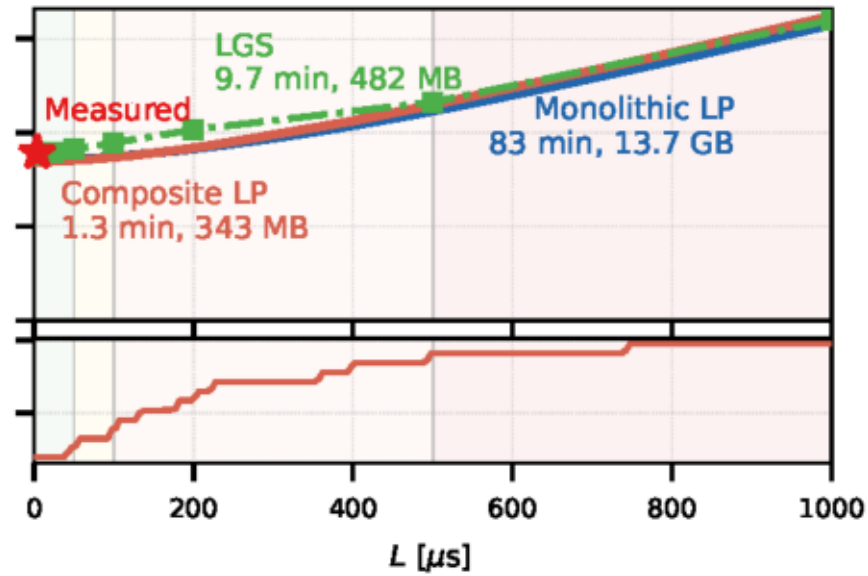
Raw-derived result: selected job 1808340 campaign



Maximum relative errors: 0.142% for latency and 0.242% for bandwidth. The matching campaign conflicts with the message-size range stated in the paper.

Figure 5: Llama training iteration

Paper plot



Closest recovered raw-derived Composite curve

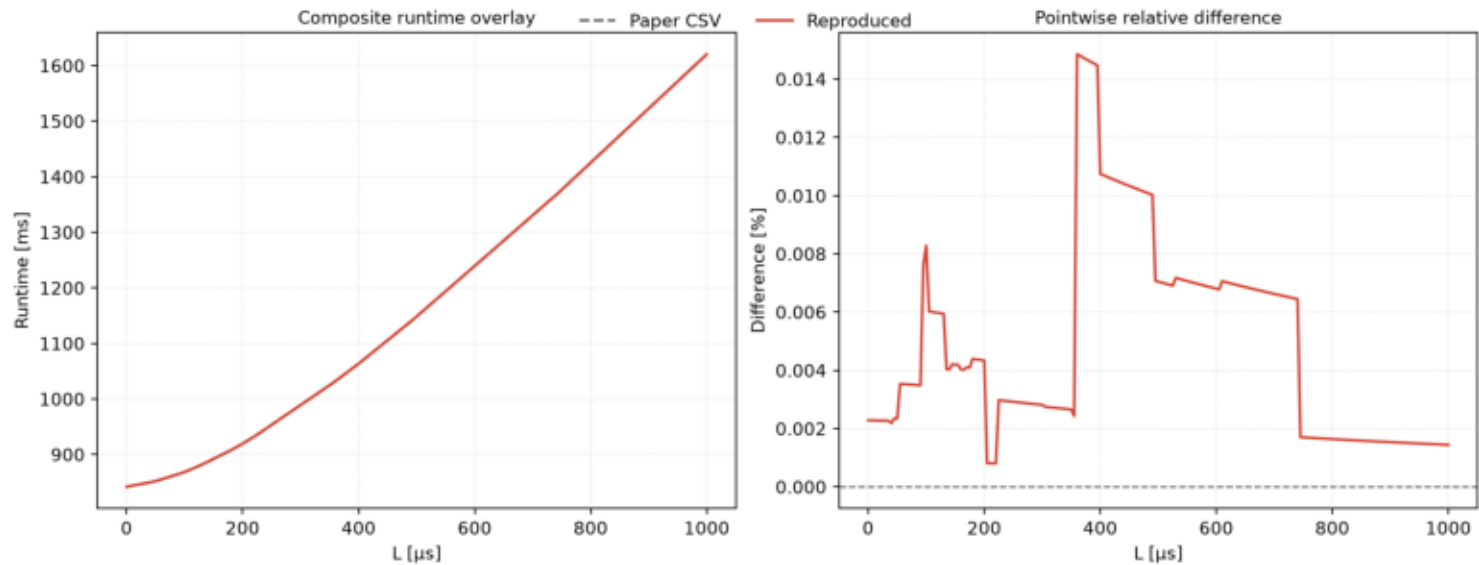
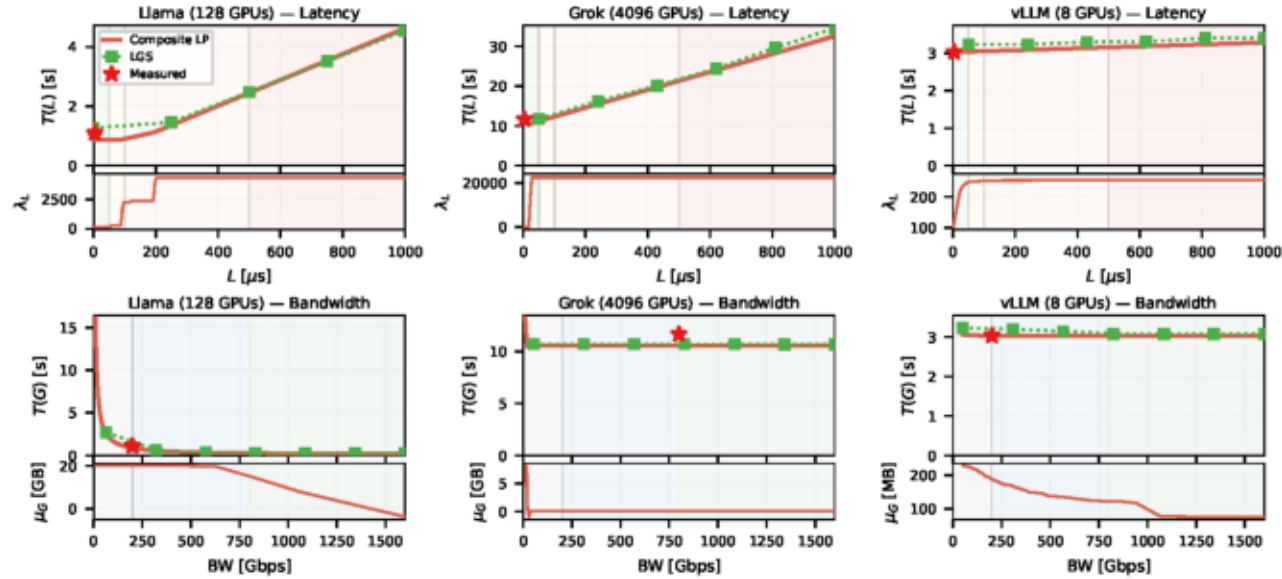
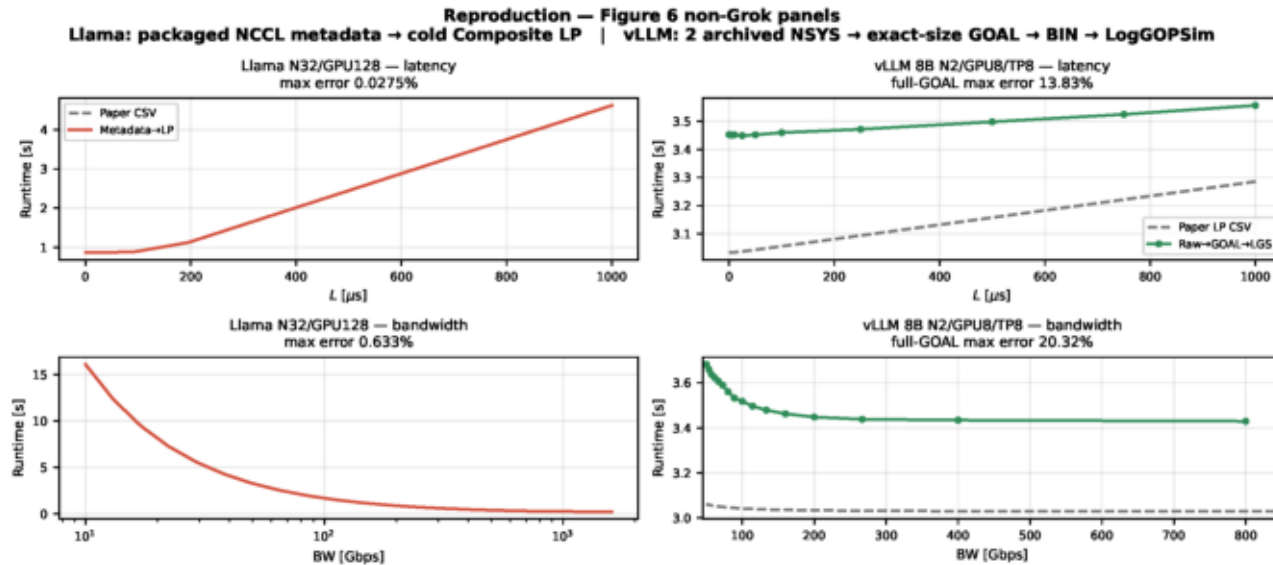


Figure 6: latency and bandwidth sensitivity

Paper plot

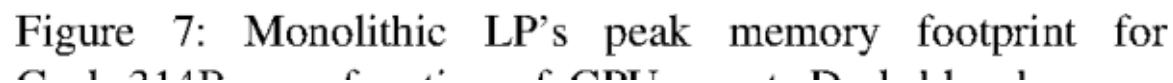


Reproduction evidence: Llama and vLLM panels



Llama uses the strongest retained metadata rerun. The archived vLLM job point replays exactly, but the full curve differs. Grok 4k uses existing CSV data and was not rerun.

Paper plot

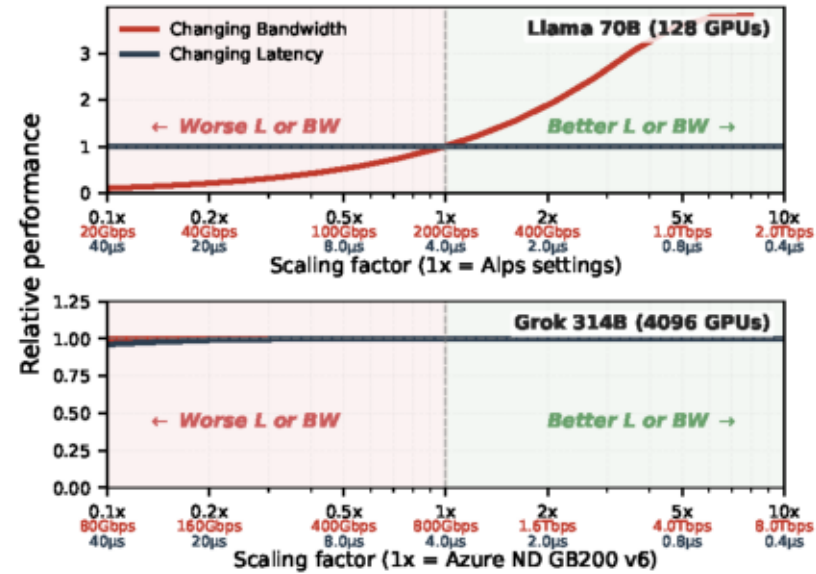


The chart displays Peak Memory [GB] on a logarithmic scale (from 10^1 to 10^5) against the Number of Nodes (N4, N8, N16, N32, N64, N128, N256, N512, N1024). The legend indicates that dark blue bars represent 'Measured' data and light blue bars represent 'Extrapolated' data. A red dashed line with circular markers shows the trend for Monolithic LP, while the bars represent Composition LP. A callout box highlights the memory difference at N1024: Monolithic LP needs ~34 TB, while Composition LP needs ~0.2 TB, which is 170 times less memory.

Number of Nodes	Nodes (decimal)	Monolithic LP Peak Memory [GB]	Composition LP Peak Memory [GB]
N4	(16)	8 GB	8 GB
N8	(32)	10 GB	10 GB
N16	(64)	22 GB	22 GB
N32	(128)	59 GB	59 GB
N64	(256)	146 GB	146 GB
N128	(512)	389 GB	389 GB
N256	(1024)	1.7 TB	~0.2 TB
N512	(2048)	7.6 TB	~0.2 TB
N1024	(4096)	34.0 TB	~0.2 TB

Figure 8: network-parameter sensitivity

Paper plot



Artifact output: published-value redraw

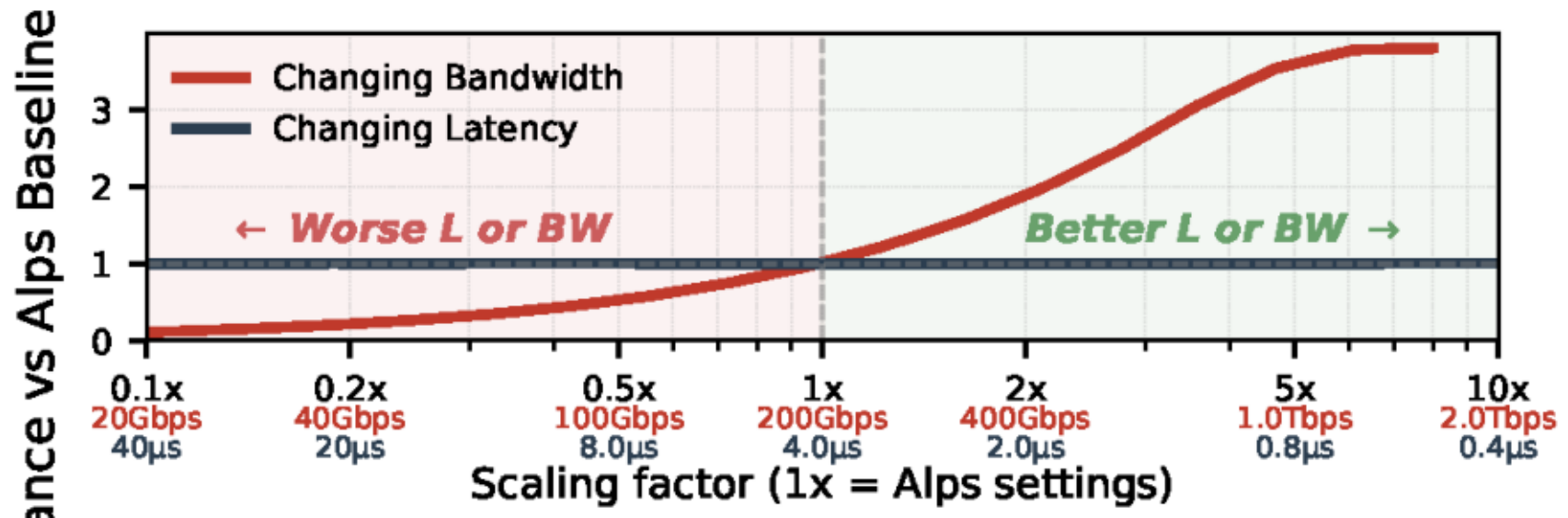
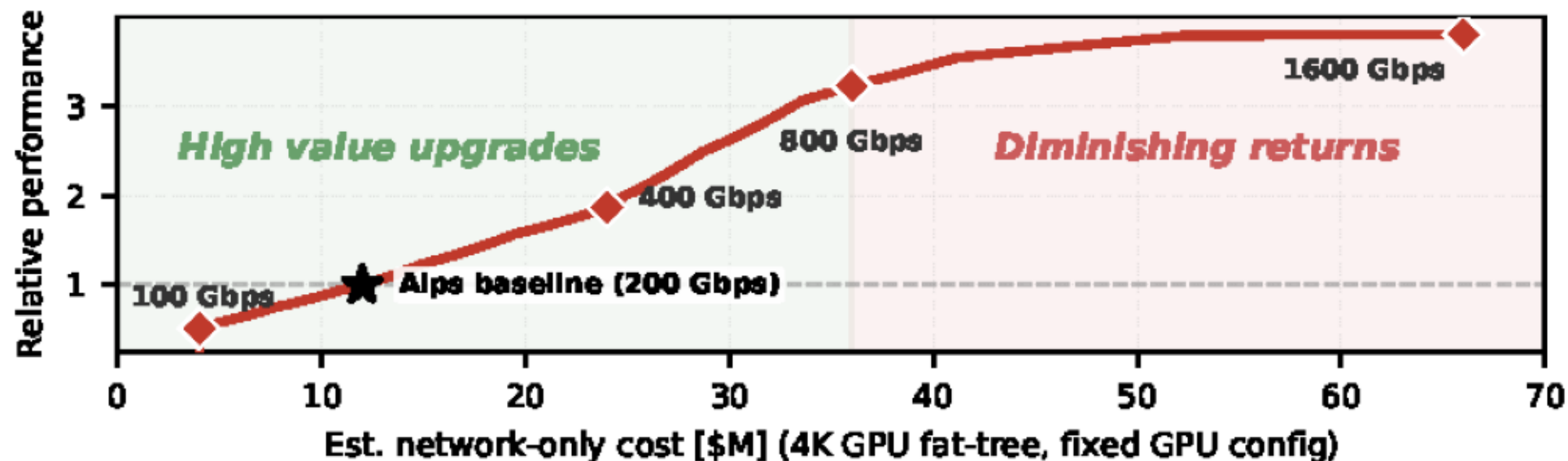


Figure 9: network cost trajectory

Paper plot



Artifact output: published-value redraw

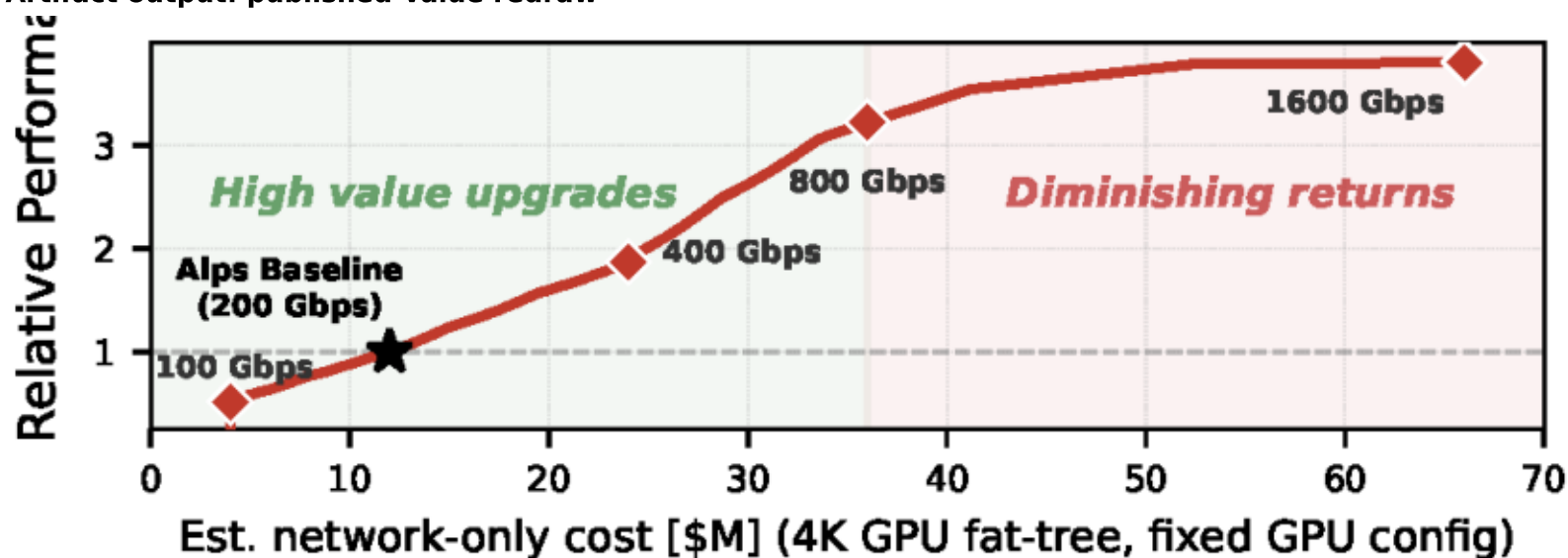
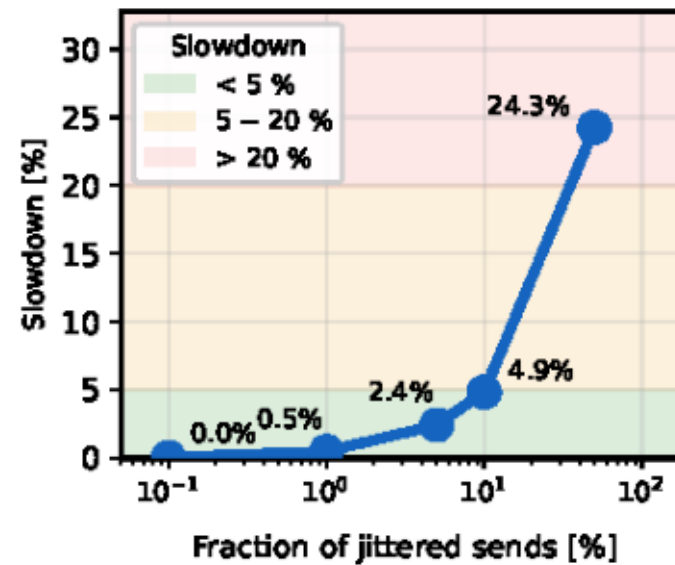
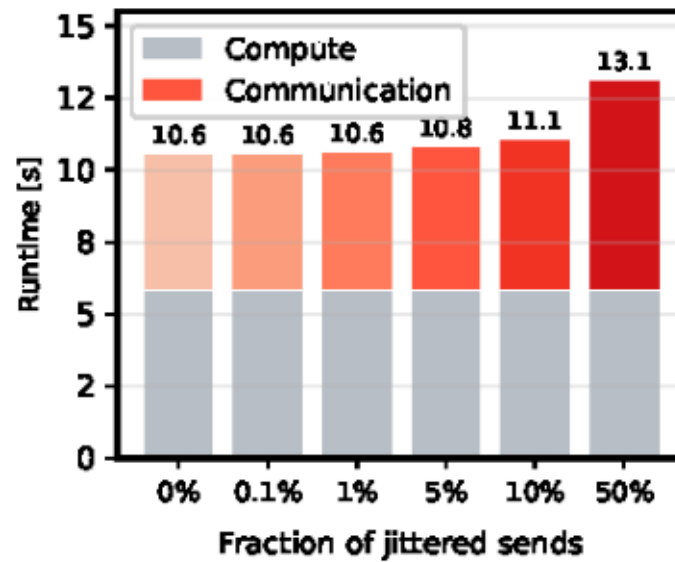


Figure 10: jitter sensitivity

Paper plot



Artifact output: compact-data redraw

